

REVIEW



Genomic tests to guide management of breast cancer in Europe: regulation, reimbursement, adoption, and challenges

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ABSTRACT

The integration of genomic tests such as the Oncotype DX Breast Recurrence Score® test, into routine clinical practice represents a significant advance in personalized breast cancer care. By supporting more tailored therapeutic decisions, these diagnostics can improve patient outcomes, while reducing risks of undertreatment, overtreatment, and associated side effects. Cost-effectiveness has already been demonstrated in numerous publications. However, for widespread adoption across Europe, four principal challenges must be overcome: regulation, technology assessment, reimbursement, and gaps in real-world evidence. Since May 2022, the European Union In Vitro Diagnostics Regulation (IVDR) has updated requirements for demonstrating clinical utility and analytical and scientific validity, creating new barriers for manufacturers regarding primary evidence generation. Variability in health technology assessment (HTA) frameworks and reimbursement mechanisms across countries further complicates adoption. Demonstrating real-world benefits of these technologies requires robust, representative data collections, yet current clinical trial evidence often underrepresents certain patient populations, raising equity concerns. Whilst the IVDR will help standardize regulatory requirements, challenges remain in harmonizing evidence standards for HTA and reimbursement. This review explores these barriers using the Oncotype DX® test as an exemplar. Evidence was drawn from targeted literature searches and reviews of regulatory, reimbursement, and gray literature relevant to European healthcare systems.

PLAIN LANGUAGE SUMMARY

Genomic tests can analyze the genes in breast tumors to help doctors better understand how the cancer is likely to behave. They provide important information to guide treatment, such as whether chemotherapy, the standard treatment used to target fast-growing cancer cells, is likely to be beneficial. This helps ensure that patients receive the most appropriate care, offering treatment when it is likely to help, and avoiding it when it is not needed. In many parts of Europe, these tests are included in national guidelines and covered by health systems, which supports broad access for patients. However, in other countries, approval and funding processes are still evolving. This creates differences in how widely the tests are used, and some patients may not have the same opportunity to benefit from personalized treatment. A key reason for this variability is that each country has its own approach to approving and reimbursing new medical technologies. The process can be complex and often requires large amounts of evidence, which may differ from one country to another. Although the tests are supported by strong scientific research, differences in national systems make it challenging to introduce them across Europe in a consistent way. Greater coordination across countries could help ensure more equal access to tumor profiling. By working toward more aligned approval and evaluation processes, health systems could support wider use of personalized testing, leading to more informed treatment decisions and improved care for people with breast cancer.

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1. Introduction

Breast cancer affects an estimated 1 in 11 women under 75 in Europe and is the third leading cause of cancer-related deaths [1,2]. While advances in surgical treatment and radiotherapy have improved survival rates, patients with early-stage disease, defined as cancer confined to the breast, with or without regional lymph node involvement, and no distant metastases, remain at risk of

recurrence [3]. Hormone-receptor-positive (HR-positive), human epidermal growth factor receptor 2-negative (HER2-negative) tumors, which account for >70% of cases, are the most common form [4].

Adjuvant chemotherapy is used to reduce recurrence risk, yet up to 70% of breast cancer patients receive little to no benefit [5], exposing them to short- and long-term toxicities such as

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Article highlights

- Genomic testing, such as the Oncotype DX test, enables personalized treatment in early-stage hormone-receptor-positive (HR-positive), human epidermal growth factor receptor 2-negative (HER2-negative) breast cancer, reducing unnecessary chemotherapy and associated toxicities.
- Despite clinical utility, adoption across Europe is inconsistent due to regulatory delays under the In Vitro Diagnostics Regulation (IVDR), fragmented reimbursement pathways, and local implementation barriers.
- Health Technology Assessment (HTA) processes vary widely, with differences in evidence requirements, manufacturer involvement, and reimbursement outcomes across European Union (EU) member states.
- New EU regulations aim to harmonize HTA through Joint Clinical Assessments, but molecular diagnostics are excluded until 2030, prolonging reliance on national systems.
- To improve access, coordinated efforts are needed, including standardized reimbursement processes, investment in laboratory infrastructure, and expanded real-world evidence generation

immunosuppression with or without infection, nausea, fatigue, alopecia, chronic fatigue, and neurocognitive impairment [6]. Other side effects such as cardiovascular damage, secondary malignancies and neuropathy can be irreversible and severely limit the quality of life. Moreover, in rare cases, complications like febrile neutropenia can be fatal. Importantly, many breast cancer patients receive chemotherapy unnecessarily due to the limitations of current clinical predictors that are used to select patients [7]. These conventional methods often lack precision, leading to overtreatment, avoidable side effects, and unnecessary costs to healthcare systems. Therefore, it is imperative to find safe ways of tailoring treatment to patients' individual needs based on tumor biology, best available evidence, and corresponding recommendations and guidelines.

Genomic testing in breast cancer, specifically tumor profiling such as the Oncotype DX Recurrence Score® test, offers a potential solution by providing a more personalized approach to treatment. By evaluating the expression of multiple genes within a breast tumor, these tests generate a risk profile specific to each patient, helping clinicians decide whether the benefits of adjuvant chemotherapy outweigh the risks [7]. In addition, emerging approaches, such as the ADAPT trial concept, suggest that breast cancer gene expression testing could be combined with additional biomarkers, such as dynamic Ki-67 during preoperative endocrine therapy, to further refine treatment strategies [8]. However, despite the clear benefits and clinical utility of breast cancer genomic testing, significant challenges remain in ensuring consistent adoption and accessibility across European healthcare systems. Barriers to adoption included the need for robust evidence generation for complex technologies [9], navigating complex regulatory and reimbursement pathways, and reimbursement variability across countries.

In this review, we aim to explore the factors influencing the adoption of breast cancer genomic test technologies, using the Oncotype DX test as an exemplar. Evidence for this review was gathered through targeted searches of PubMed and Google Scholar using keywords related to breast cancer, tumor profiling, and health technology assessment. These

searches were supplemented by reviews of regulatory and reimbursement agency websites and relevant gray literature. References were selected to provide a comprehensive overview of regulatory, reimbursement, and adoption considerations within European healthcare systems. We assess these regulatory frameworks, reimbursement mechanisms, and local clinical practices across key European countries, and highlight the inconsistencies and challenges in the integration of these technologies into routine clinical practice. Finally, we discuss strategies to overcome existing barriers and enhance the adoption of these valuable tools with the goal of optimizing breast cancer management.

2. Journey: from inception to adoption

Genomic testing has transformed the landscape of breast cancer management by enabling more personalized treatment decisions [10–13]. The development and implementation of these technologies can be envisaged as a journey, with several barriers needing to be overcome before the technologies are widely adopted and individuals and healthcare organizations reap their benefits. The journey is illustrated schematically in Figure 1.

2.1. Fundamental research

The foundation of genomic test technologies lie in extensive clinical research focused on demonstrating their analytical and clinical validity. The Oncotype DX test, for instance, was developed to assess the relative expression of 21 genes, generating a Recurrence Score® result that informs the need for adjuvant chemotherapy in early-stage breast cancer patients [10]. This research was conducted in specialized molecular oncology laboratories in the United States, where rigorous experimental validation, assay development, and clinical validation studies were performed under strict quality control standards. The adherence to high laboratory and regulatory standards during the research phase was critical in ensuring the test's reliability and eventual integration into clinical practice. Alongside the Oncotype DX test, other genomic assays such as EndoPredict, Prosigna, MammaPrint, and the Breast Cancer Index have been developed to guide treatment decisions in early breast cancer. These tests differ in gene panels, methodology, and whether they are performed centrally or locally, but all aim to refine recurrence risk assessment and optimize therapy selection. Their coexistence highlights both the progress in genomic profiling and the challenges of integrating multiple platforms with varying evidence bases into routine practice. A summary of key attributes of the Oncotype DX test and related technologies is reported in Table 1.

2.2. Regulatory pathways

While these technologies have demonstrated significant clinical value through rigorous research, this represents only the start of their journey from bench to clinic. Achieving regulatory approval remains a critical step, where navigating the complex frameworks and varying requirements across European markets can significantly impact the adoption

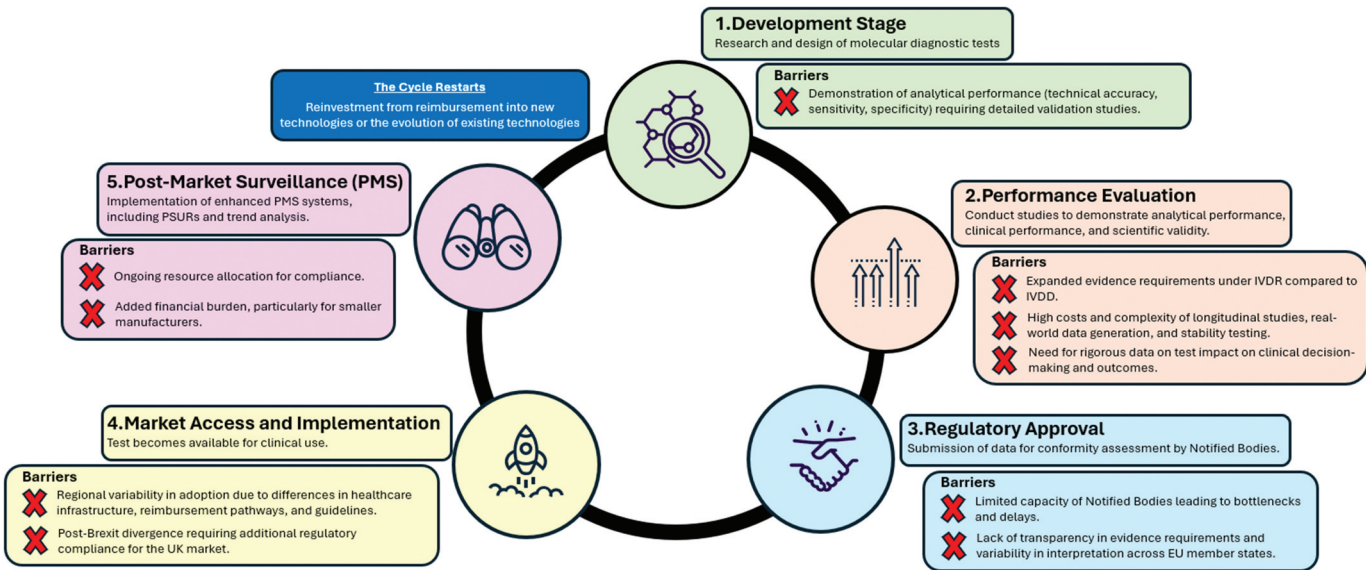


Figure 1. Regulatory pathway for molecular diagnostics, including barriers.
Abbreviations: IVDD, In Vitro Diagnostic Directive; IVDR, In Vitro Diagnostic Regulation; PMS, post-market surveillance.

Table 1. Comparison of genomic test attributes.

Attribute/Element	Genomic Test Technology				
	Oncotype DX [10,14]	EndoPredict [11,15]	Prosigna [12,16]	MammaPrint [13,17]	Breast cancer index [18,19]
Number of Genes	21	12	50	70	11
Prognostic or predictive	Both; predictive of chemotherapy benefit	Prognostic	Prognostic	Prognostic	Both; predictive of benefit from extended ET
Type of Risk Output	Recurrence Score (0–100 scale)	EPclin risk score (low vs high risk)	Risk of recurrence score (ROR)	Binary risk score (low vs high risk)	Risk of recurrence 0–10 and 5–10 years after diagnosis
Methodology	RT-qPCR	RT-qPCR	Nanostring nCounter technology	Microarray	RT-qPCR
Tissue Requirement	FFPE tissue	FFPE tissue	FFPE tissue	FFPE or fresh tissue	FFPE tissue
Assessment	Central (US/Germany)	Local	Local	Central (Netherlands/US)	Central (US)

Abbreviations: EPclin, EndoPredict clinical risk score; ET, endocrine therapy; FFPE, Formalin-fixed paraffin-embedded; RT-PCR, Reverse transcription polymerase chain reaction; RT-qPCR, Reverse transcription quantitative polymerase chain reaction.

timeline. One of the most significant developments in recent years has been the introduction of the In Vitro Diagnostic Regulation (IVDR), replacing the previous In Vitro Diagnostic Directive (IVDD), to address several shortcomings [20]. As diagnostic technology has advanced, the IVDD failed to adequately account for innovations such as genomic testing, next-generation sequencing, and other molecular diagnostics. The IVDR aims to establish a robust, transparent, predictable, and sustainable regulatory framework for *in vitro* diagnostic medical devices [21], to enhance patient safety, improve the reliability of *in vitro* diagnostics, and ensure that devices meet rigorous standards of clinical performance and analytical validity [20]. However, these goals come at the cost of heightened requirements and increased regulatory timelines, as companies are required to provide robust data on both the clinical and analytical performance of their devices, often beyond what was previously necessary under the IVDD [22].

The transition to the IVDR initially provided a five-year period for manufacturers to adapt to the new regulatory framework [23]. As the initial deadline approached, it became clear that the combination of heightened regulatory demands,

the expanded scope of diagnostics requiring certification, and disruptions caused by the COVID-19 pandemic had left many manufacturers and Notified Bodies unprepared. Despite an adjustment by the European Parliament and Council, ongoing concerns regarding the limited capacity of Notified Bodies to undertake conformity assessments led to a further extension. This decision aimed to prevent shortages of critical devices, but crucially highlights a limited capacity of accredited Notified Bodies capable of conducting conformity assessments under the more stringent requirements of IVDR, compared to IVDD. Ultimately, for manufacturers of advanced diagnostics like the Oncotype DX test, this translates into bottlenecks, longer timelines for regulatory approval, higher costs, and delays in bringing innovative products to market [23].

The more stringent requirements of IVDR compared to IVDD begin during performance evaluation. Unlike its predecessor, the IVDR mandates a structured and comprehensive evaluation process for diagnostic technology that includes three critical components, namely analytical performance, clinical performance, and scientific validity [22]. A summary comparing IVDD and IVDR is shown in Table 2. While these changes aim to ensure that devices are safe, reliable, and

Table 2. Summary of key differences between IVDD and IVDR.

Aspect	IVDD	IVDR
Analytical Performance	Limited focus, minimal documentation	Mandatory, detailed technical assessment
Clinical Performance	Often optional, limited to certain devices	Mandatory for all but the lowest-risk devices
Scientific Validity	Assumed or inferred, little evaluation	Explicitly required, systematic evaluation
Documentation	Minimal reporting requirements	Detailed performance evaluation reports
Regulatory Oversight	Limited involvement of Notified Bodies	Comprehensive scrutiny by Notified Bodies

Abbreviations: IVDD, In Vitro Diagnostic Directive; IVDR, In Vitro Diagnostic Regulation.

clinically meaningful, they also present considerable barriers to market access.

Under the IVDR, analytical and clinical performance requires manufacturers to demonstrate the technical accuracy, precision, sensitivity, specificity, and reproducibility in laboratory settings, as well as meaningful results in real-world settings [20,24]. With increasing device complexity, such as with high-risk devices like the Oncotype DX test, a higher volume of studies demonstrating analytical and clinical performance are required. Hence, even minor gaps during the analytical performance assessment could trigger the need for the generation of new or additional evidence, resulting in delays in market access [24].

Compounding these issues is the lack of a single, clear, and accountable structure governing the oversight of the IVDR framework [21]. Upon completion of the transition period, the IVDR strives to be a more harmonized regulatory framework, consistent across EU countries (excluding the UK post Brexit) [20]. However, governance is shared among 27 national Competent Authorities, the Medical Device Coordination Group (MDCG), the European Commission, and individual National Competent Authorities responsible for supervising Notified Bodies [21].

2.3. Reimbursement pathways

In parallel with regulatory oversight, another key consideration for successful market access is reimbursement. Following regulatory approval, reimbursement pathways are usually the next critical step in the process of medical technology adoption. In most European countries, this is done through national Health Technology Assessment (HTA) bodies, which evaluate new technologies for clinical effectiveness, cost-effectiveness, and overall impact on healthcare systems. While the IVDR ensures regulatory compliance by standardizing safety and performance criteria across Europe, the approach to HTA and reimbursement has historically varied across member states. For example, in Ireland, the Oncotype DX test was available to clinicians from 2007 but did not gain HTA approval until 2011, with reimbursement only granted in 2013, reflecting a staged and protracted pathway [25,26]. Germany exemplifies how such regulatory hurdles can eventually be overcome. After comprehensive assessment by Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (IQWiG) and the Gemeinsamer Bundesausschuss (GBA), the Oncotype DX test received tariff assignment in 2021, enabling nationwide reimbursement and illustrating how rigorous processes, once completed, can translate into predictable access [27–29]. In Italy, regional reimbursement began in 2019, but national reimbursement was only achieved in 2021, reflecting longer timelines

and marked regional variation prior to central approval [30]. Together, these contrasting experiences highlight the diversity in how European member states approach HTA and reimbursement.

2.4. HTA submission processes and their impact on companies

Despite their common goal of assessing new technologies, HTA processes differ significantly across Europe, particularly in terms of how evaluations are conducted, and the role manufacturers play in the process, even within their own authorities. In some countries, companies submit full evidence dossiers and actively participate in assessments, while in others, evaluations are conducted independently of the manufacturer, limiting their influence over the outcome. For example, in England, National Institute for Health and Care Excellence (NICE) evaluates diagnostics through two distinct programs, which differ in how much control manufacturers have over the assessment. For instance, the Diagnostics Assessment Programme (DAP) does not mandate manufacturers to present their own evidence submission, but commissions an External Review Group (ERG) to conduct independent clinical- and cost-effectiveness evaluation. Whilst this limits the influence of manufacturers in the process, it can also reduce their administrative burden [31]. In contrast, the Medical Technologies Evaluation Programme (MTEP) requires manufacturers to submit full evidence dossiers, allowing them to provide their own clinical and economic data, but this may require significant financial investment [32], which may disproportionately impact smaller companies.

In contrast, other European HTA agencies, such as the Norwegian Medicines Agency (NoMA) in Norway and Red Española de Agencias de Evaluación de Tecnologías Sanitarias y Prestaciones del Sistema Nacional de Salud (RedETS) in Spain, allow companies to submit applications and participate directly in the assessment process. This means manufacturers have greater control over how their test is evaluated and can engage in discussions over cost-effectiveness modeling and clinical utility assessments. The variation in HTA submission models across Europe adds another layer of complexity for companies developing genomic tests, as they must navigate both financial and strategic considerations when entering different markets. In France, Haute Autorité de Santé (HAS) issued an unfavorable assessment of multigene signatures including the Oncotype DX test in 2019, reaffirmed by a cautious 2023 update, which restricted broad reimbursement despite international guideline support [33,34]. In England, NICE approved tumor profiling through its Diagnostics Assessment Programme in 2013,

but because HTA recommendations are not mandatory, national NHS adoption lagged until 2015, with persistent variability in uptake [25,35–37]. By contrast, in Norway, a 2023 NIPH assessment was rapidly followed by a national decision in 2024 to introduce the Oncotype DX test, exemplifying how centralized HTA can accelerate patient access [38,39]. Together, these experiences illustrate the wide divergence in HTA submission processes and their impact on adoption timelines.

2.5. The role of regulation (EU) 2021/2282 in standardising HTA processes

Up until 12 January 2025, when the voluntary adoption of Health Technology Assessment Reports (HTAR) and Joint Clinical Assessments (JCA) for certain health technologies across EU member states took effect under Regulation (EU) 2021/2282 [40], significant discrepancies existed in how HTA processes were implemented, outlined in Table 3. Previously, technology developers and manufacturers were required to generate country-specific evidence to demonstrate the economic benefits of implementing new diagnostic and prognostic tests [45]. Required evidence varies considerably between countries – for example, Ireland emphasizes analytical validity and cost-effectiveness [25,26], whereas France requires full economic evaluation [46], and Germany demands demonstration of patient-relevant benefit and clinical utility [29]. These requirements posed significant challenges, hindering equitable reimbursement and delaying timely access to diagnostic testing across the continent [33].

In response, HTAR provides a structured framework for evaluating new technologies across the EU, focusing on clinical effectiveness and cost-effectiveness, while JCA establishes a centralized process ensuring that all EU member states assess new diagnostics based on the same clinical evidence, thereby reducing duplicative efforts. However, molecular diagnostics will not fall within the scope of JCA assessments until 2030 [40], meaning developers of these technologies must continue engaging with national HTA bodies for reimbursement decisions. Furthermore, these novel frameworks are designed to complement, not replace, national HTA processes [47]. Countries will still retain authority over reimbursement decisions, and national bodies, such as the Haute Autorité de Santé (HAS) in France and Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (IQWiG) in Germany, will continue to assess cost-effectiveness and budget impact at the national level. As a result, companies seeking reimbursement for medical and diagnostic devices must still navigate these nuanced national pathways, ensuring compliance with country-specific evidence and economic evaluation requirements.

At the time of publication, it remains to be seen whether these harmonization efforts will successfully streamline reimbursement processes and improve access to diagnostic technologies. Challenges may arise in the integration of JCA findings into national reimbursement frameworks, potential delays due to administrative adjustments, and differences in budget allocation across member states.

As shown in Table 4, differences in scope, mandates, and funding requirements across HTA agencies influence how diagnostics are assessed and adopted. In countries like France and Germany, HTA processes for diagnostics are mandatory. In France, a positive recommendation typically enables reimbursement, whereas in Germany, reimbursement is generally guaranteed once a diagnostic is approved, with the HTA outcome informing the conditions and level of reimbursement. These agencies, such as HAS and IQWiG, require extensive clinical and economic evidence, including real-world data, to demonstrate value [29,46]. Although this structured approach provides clarity, the evidentiary burden can delay market entry, particularly for smaller manufacturers with limited resources. In contrast, non-mandatory HTA processes in Ireland and England create variability and uncertainty. For instance, in England, NICE evaluates diagnostics through its Diagnostics Assessment Programme (DAP) and Medical Technologies Evaluation Programme (MTEP), but the lack of a mandatory HTA pathway means that even positive recommendations do not guarantee adoption across the NHS [37]. Similarly, in Ireland, HTA evaluations by the Health Information and Quality Authority (HIQA) are not required, and funding is not automatically secured, leaving the uptake of diagnostics dependent on other factors [48].

2.6. Healthcare funding

While healthcare services have evolved differently across Europe, the complexities of country-specific healthcare funding models are beyond the scope of this review. However, they share a common foundation: all are socialized models designed to ensure access based on medical need rather than ability to pay, whether funded through direct taxation or social insurance. As a result, these variations in funding mechanisms are unlikely to significantly impact the adoption of new technologies such as the Oncotype DX test, particularly given its role as an essential part of the patient pathway rather than an optional extra. The key issue instead may arise from the way diagnostics are funded compared to where their benefits are realized in system utilizing siloed budgeting practices. For instance, in many systems, laboratories or local budgets bear the upfront costs of diagnostic tests, while the savings and clinical benefits, such as reducing unnecessary treatments or hospitalizations, are accrued elsewhere in the healthcare system [54]. This separation of cost and benefit means that the entities responsible for funding diagnostics often lack a direct financial incentive to adopt them, slowing their integration into routine care.

2.7. Aligning reimbursement pathways

While the alignment of reimbursement pathways across European markets would streamline access to innovative diagnostics, the diverse healthcare models and the sovereignty retained by individual countries poses many challenges. Despite the singular approach for achieving regulatory

Table 3. Comparison of HTA requirements across European countries.

Country	Regulatory Authorities	HTA Agency	Required Evidence	Timeframe for Approval (Oncotype DX)	External Quality Assessment Mandate
Ireland	EMA/HPRA	HIQA	- Analytical validity - Clinical effectiveness (validity & utility) - Comparative effectiveness - Cost-effectiveness	- Became available: 2007 - HTA Approved: 2011 - Approved for reimbursement: 2013 [25,26]	Yes
Spain	EMA/AEMPS	RedETS	- Clinical effectiveness (validity & utility) - Safety - Comparative effectiveness - Cost-effectiveness - Ethical, legal & social aspects - Organisational impact	- HTA Approved: 2014 - Regional reimbursement (non-centralized decision): since at least 2017 [41,42]	No mandate, regional variation
England	MHRA	NICE	- Analytical validity - Clinical effectiveness (validity & utility) - Comparative effectiveness - Cost effectiveness	- Became available: 2007 - HTA Approved (UK wide): 2013 - Full Implementation (NHS access): 2015 [25,35,36]	Yes
Scotland	MHRA	SMC	- Safety & credibility - Clinical effectiveness (validity & utility) - Comparative effectiveness - Cost effectiveness	- Became available: 2007 - HTA Approved (UK wide): 2013 - Formal endorsement (MPEP): 2016, 2018 - Evidence updated: 2023 [25,35,36,43]	Yes
Norway	NoMA	NIPH	- Safety - Clinical effectiveness (validity & utility) - Cost-effectiveness - Budget impact	- HTA Approved: 2023 - Approved for reimbursement: 2024 - Full Implementation: Expected [38,39]	Yes
Italy	EMA	AGENAS	- Clinical effectiveness (validity & utility) - Safety - Economic Evaluation - Ethical, legal & social aspects - Organisational impact	- First regional reimbursement (non-centralised decision): 2019 - National reimbursement: 2021 [30]	No mandate, regional variation
France	EMA/ANSM	HAS	- Clinical effectiveness (validity & utility) - Economic evaluation	- Became available: unclear, likely pre-2015 - Unfavorable HTA assessment (along with MammaPrint, Prosigna, EndoPredict): 2019 - No full reimbursement: 2023 [34]	Yes
Germany	EMA/BfArM	IQWiG/GBA	- Patient-relevant benefit - Diagnostic accuracy - Clinical utility - Safety - Economic evaluation	- HTA Approved: 2019 - Full Implementation (EBM tariff assigned): 2021 [27,28]	Yes
Poland	EMA/URPL	AOTMiT	- Clinical effectiveness (validity & utility) - Safety - Economic Analysis - Impact on Healthcare System	- real-life clinical utility research: 2021–22 (published 2024) - HTA Approved: Not yet received [44]	No mandate, voluntary participation

Abbreviation: AEMPS, Agencia Española de Medicamentos y Productos Sanitarios (Spanish Agency for Medicines and Health Products); AGENAS, Agenzia Nazionale per i Servizi Sanitari Regionali (National Agency for Regional Health Services); AIFA, Agenzia Italiana del Farmaco (Italian Medicines Agency); ANSM, Agence Nationale de Sécurité du Médicament et des Produits de Santé (National Agency for the Safety of Medicines and Health Products); AOTMiT, Agencja Oceny Technologii Medycznych i Taryfikacji (Agency for Health Technology Assessment and Tariff System); BfArM, Bundesinstitut für Arzneimittel und Medizinprodukte (Federal Institute for Drugs and Medical Devices); EBM, Einheitlicher Bewertungsmaßstab (Uniform assessment criteria); EMA, European Medicines Agency; GBA, Gemeinsamer Bundesausschuss (Federal Joint Committee); HAS, Haute Autorité de Santé (French National Authority for Health); HIQA, Health Information and Quality Authority; HPRA, Health Products Regulatory Authority; HTA, Health Technology Assessment; IQWiG, Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (Institute for Quality and Efficiency in Health Care); MPEP, Molecular Pathology Evaluation Panel; NHS, National Health Service; NICE, National Institute for Health and Care Excellence; NIPH, Norwegian Institute of Public Health; RedETS, Red de Evaluación de Tecnologías Sanitarias (Health Technology Assessment Network); SHTG, Scottish Health Technologies Group; UK, United Kingdom; URPL, Urząd Rejestracji Produktów Leczniczych (Office for Registration of Medicinal Products).

Table 4. Overview of HTA Agency processes and funding requirements for diagnostic technologies across selected European countries.

Country	HTA Agency	Funding	Scope Includes Diagnostics	Mandatory HTA Process	Mandatory Funding After Positive HTA
Ireland [48]	HIQA	Centrally	Yes	No	No
Spain [49]	RedETS	Locally	Yes	Yes	Yes
England [37]	NICE	Centrally	Yes	No	No
Scotland [50]	SHTG	Centrally	Yes	No	No
Norway [51]	NIPH	Centrally	Yes	Yes	Yes
Italy [52]	AGENAS	Locally	Yes	Yes	Yes
France [46]	HAS	Centrally	Yes	Yes	Yes
Germany [29]	IQWiG/GBA	Centrally	Yes	Yes	Yes
Poland [53]	AOTMiT	Centrally	Yes	Yes	No

Abbreviations: AGENAS, Agenzia Nazionale per i Servizi Sanitari Regionali (National Agency for Regional Health Services); AOTMiT, Agencja Oceny Technologii Medycznych i Taryfikacji (Agency for Health Technology Assessment and Tariff System); GBA, Gemeinsamer Bundesausschuss (Federal Joint Committee); HAS, Haute Autorité de Santé (French National Authority for Health); HIQA, Health Information and Quality Authority; HTA, Health Technology Assessment; IQWiG, Institut für Qualität und Wirtschaftlichkeit im Gesundheitswesen (Institute for Quality and Efficiency in Health Care); NICE, National Institute for Health and Care Excellence; NIPH, Norwegian Institute of Public Health; RedETS, Red de Evaluación de Tecnologías Sanitarias (Health Technology Assessment Network); SHTG, Scottish Health Technologies Group.

authorization in Europe, the reality is that reimbursement will likely always require a market-specific approach.

3. Local implantation and adoption challenges

The successful adoption of molecular diagnostics, such as biomarker tests and next-generation sequencing panels, hinges on addressing practical considerations at the local level, ensuring robust post-marketing surveillance and leveraging the influence of Key Opinion Leaders (KOLs) and local champions [55–57]. Differences in laboratory infrastructure, resource availability, and clinical guidelines across Europe determine how quickly and effectively molecular diagnostics can be implemented [33]. Strengthening local capacity through investment in laboratory training programs, standardized protocols, and quality assurance systems is essential for ensuring consistent and reliable results that clinicians trust [57], with high-quality evidence, such as that from the TAILORx clinical trial [5], playing a crucial role in building confidence and driving adoption.

KOLs and local champions significantly influence the adoption of diagnostics. KOLs can drive the appetite for new technologies by educating their peers about the clinical and economic benefits of diagnostics, shaping treatment guidelines, and addressing skepticism among clinicians and decision-makers [55]. Local champions, who often have strong ties to specific healthcare systems or institutions, bridge the gap between national policies and regional implementation [58]. Their ability to align adoption efforts with local priorities and address context-specific challenges is invaluable in overcoming resistance to change.

3.1. Post market surveillance

Post market surveillance (PMS), which has become more rigorous under the IVDR [59], also plays a critical role in ensuring the real-world clinical utility of breast cancer genomic tests is evaluated and realized [56]. While PMS primarily focuses on regulatory compliance, mandating periodic safety update reports (PSURs), trend reporting and systems for managing serious incidents, these processes also generate critical real-world evidence that supports adoption. By validating test performance across diverse clinical settings, PMS provides

additional data to build clinician and payer confidence, address variability in test outcomes, and refine protocols [20].

Continual improvement of diagnostic assays in different clinical settings ensures their long-term clinical relevance and reliability. As new biomarkers are discovered and technologies advance, molecular diagnostics must evolve to maintain their value in guiding personalized treatment decisions. Efforts to expand biomarker panels, refine analytical methods, and integrate advanced tools, such as machine learning, are crucial for enhancing test accuracy and interpretability [60]. These advancements help address clinician concerns and strengthen confidence in the clinical utility of molecular diagnostics. A multifaceted approach that combines local adaptation, ongoing validation, and stakeholder engagement is essential for optimizing the adoption of diagnostics across diverse healthcare systems.

Finally, while the generation of real-world data through post-market surveillance is vital for validating test utility and supporting adoption, questions remain about how such research should be funded. Whether responsibility lies with public institutions, payers, industry, or academic bodies is still debated, and clearer frameworks may be needed to ensure the sustainability of health services research in this space.

3.2. From trials to the real-world

The widespread adoption of molecular profiling technologies in cancer patients in translational research and clinical practice faces numerous practical challenges. Under the new IVDR framework, there is an increased focus on demonstrating the clinical utility, analytical validity, and cost-effectiveness of diagnostic tests [20]. These requirements necessitate comprehensive datasets to ensure regulatory compliance. However, large-scale, prospective, and longitudinal studies that incorporate multiple genomic analysis alongside detailed clinical histories, imaging, and pathology assessments, remain uncommon in cancer research [61]. Without these studies, regulatory approval processes may face delays, payers may hesitate to reimburse these tests, and clinicians may remain skeptical about integrating them into routine practice.

Furthermore, data collection, quality, standards, and interoperability are widely lacking in many of Europe's health and research systems, with the exception of some, like the Netherlands. There, testing standards and quality are elevated

due to biomarker testing being conducted through certified laboratories that follow fixed guidelines, strict protocols, and standardized assays [62]. Variability in testing quality and the accuracy of results, both across Europe and within individual countries between labs, undermines confidence in biomarker testing and the data generated from these tests [63].

These evidence gaps not only hinder regulatory progress and clinical uptake but also highlight deeper issues around inclusivity and representativeness in the research itself, particularly where race, ethnicity and ancestry are concerned. Ethnically diverse populations remain significantly underrepresented in precision and genomic medicine research, with growing evidence indicating that this lack of inclusion contributes to disparities in access to the genomic medicine services developed as a result [64–66]. Research has shown genetic databases, such as UK Biobank, largely include populations of European ancestry [67], with diverse ethnic groups rarely engaged, or when they are, it often seems tokenistic [68]. Consequently, individuals representing the different ethnic groups are not part of shaping or informing the focus of this area of research. Potential strategies to address these disparities include meaningful, sustained, and tailored community engagement activities across the healthcare systems, diversifying research participation, and development of Equality Diversity and Inclusion frameworks [68].

4. Perspectives

Addressing these challenges requires a comprehensive understanding of the diverse stakeholders involved in integrating breast cancer genomic testing into clinical practice. The successful integration of breast cancer genomic testing into clinical practice relies on aligning the interests and expectations of multiple stakeholders, including patients, patient advocates, providers (hospitals and clinicians), payers (healthcare systems) and developers. Each has their importance in determining the adoption, utilization, and sustainability of genomic tests [69].

For instance, patients stand to benefit directly from personalized genomic testing, which enables a more tailored approach to treatment. By accurately assessing the risk of recurrence, these technologies can help patients avoid the unnecessary physical and psychological burden of chemotherapy when it is unlikely to provide additional benefit [70]. At the same time, patients at high genomic risk can be identified early and appropriately treated, helping to prevent local relapses and metastases. Moreover, patients who receive chemotherapy based on a gene expression test often feel better informed about their treatment decisions, which can lead to greater adherence [71]. Conversely, for hospitals and healthcare institutions, while the initial investment in genomic tests may be substantial, the ability to avoid unnecessary chemotherapy can lead to downstream efficiencies, potential cost savings, and improved patient outcomes [70,72].

Increasingly, healthcare systems employ HTA agencies to decide whether genomic tests should be funded within national programs. The challenge lies in demonstrating not only the clinical utility but also the economic value of these tests to fit within constrained public health budgets while

improving patient care. However, as HTA evaluations are grounded in health economics paradigms, their decisions may not always align with the perspectives of clinicians or the broader concerns of patients.

Finally, the developers and manufacturers of the technologies are essential for generating the innovations. Ongoing research and development are needed to refine their application to specific patient cohorts and disease stages, but this requires adequate incentives and rewards to sustain progress, a balance that agencies like the National Institute for Health and Care Excellence (NICE) strive to achieve.

4.1. Future perspective

Building on the complex stakeholder dynamics and challenges outlined above, future efforts must focus on practical strategies to advance the adoption of molecular diagnostics despite the fragmented European healthcare landscape. While harmonization of HTAs across Europe, through streamlined evidence requirements and consistent reimbursement processes, would be an ideal solution to accelerate the adoption of molecular diagnostics, it remains an unrealistic goal in the current healthcare landscape. The diversity of healthcare systems, funding mechanisms, and local priorities across European countries makes such alignment highly challenging. Instead, efforts should focus on more achievable measures, such as establishing mandatory national reimbursement pathways for molecular diagnostics. Treating these technologies with the same urgency and importance as pharmaceuticals, along with clear, standardized evidence, can help drive adoption while respecting the nuances of individual healthcare systems. At the same time, Europe could work toward incremental collaboration by centralizing genomic and clinical data to generate robust cost-benefit evidence. Initiatives like the “Big Data for Better Outcomes” (BD4BO) project offer valuable frameworks for this, uniting diverse stakeholders (patients, payers, regulators) and researchers to leverage existing databases and align evidence-generation efforts [73]. While full harmonization remains unattainable, fostering data centralization and cross-border initiatives can strengthen national adoption efforts, ultimately ensuring equitable and timely access to molecular diagnostics.

5. Conclusions

Genomic test technologies like the Oncotype DX test have revolutionized breast cancer management by enabling personalized treatment strategies that optimize clinical outcomes while reducing unnecessary interventions. Despite their promise, the adoption of these diagnostics across Europe faces significant hurdles, including stringent regulatory demands under the IVDR, fragmented reimbursement pathways, and disparities in access driven by local healthcare infrastructure and policies. Addressing these challenges requires harmonized efforts to streamline regulatory and reimbursement processes, ensure consistent evidence standards, and foster collaboration among stakeholders. Initiatives to centralize genomic data and implement value-based access programs could provide the foundation for

more equitable and efficient integration of these technologies into clinical practice, such as the Big Data For Better Outcomes (BD4BO) project. By tackling these systemic barriers, the healthcare community can unlock the full potential of genomic testing, ensuring that advancements in personalized medicine benefit patients universally, reduce economic inefficiencies, and improve outcomes across diverse healthcare systems.

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